

ARACHIS ONLINE TRAINING

WHITE GLUE MANUFACTURING MODULE

Practical Small Scale Production Guide For Zimbabwe

INTRODUCTION

Welcome to the Arachis Online Training Glue Manufacturing Module.

In this lesson you will learn how to manufacture a practical and affordable white glue suitable for:

School glue
Paper glue
Carton glue
Craft glue
Light wood applications
General household use

This formula is designed for:

- * Small business startups
- * Home industries
- * Beginners learning for the first time
- * Low-cost production in Zimbabwe

The formula uses ingredients that are relatively accessible locally.

UNDERSTANDING WHITE GLUE

White glue is mainly made from **PVA (Polyvinyl Acetate)** binder.

PVA is the main ingredient responsible for:

- * sticking strength
- * flexibility
- * smooth texture
- * white appearance

Other ingredients help:

- * thicken the glue
- * preserve it
- * improve appearance
- * reduce production costs

EQUIPMENT REQUIRED

Before starting, prepare the following equipment:

Equipment	Purpose	
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25–30 litre bucket	Mixing	
Wooden stick or mixer	Stirring	
Measuring jug	Measuring liquids	
Scale	Measuring powders	
Sieve	Prevent lumps	
Bottles or containers	Packaging	
Gloves	Safety	
Mask (recommended)	Protection from powders	

INGREDIENTS — 20 LITRE BATCH

Ingredient	Quantity	Function	
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PVA Binder / PVAc Emulsion	8 litres	Main adhesive	
Clean Water	10 litres	Dilution	
CMC (Carboxymethyl Cellulose)	150 g	Thickener	
Preservative	30–50 ml	Prevents mould and bad smell	
White pigment (optional)	50 g	Improves whiteness	
Fragrance (optional)	10 ml	Better smell	
Defoamer (optional)	5–10 ml	Removes bubbles	

UNDERSTANDING EACH INGREDIENT

1. PVA Binder

This is the heart of the glue.

It provides:

- * bonding power
- * flexibility
- * smooth finish
- * durability

Without enough PVA:

- * glue becomes weak

- * paper separates easily
- * product feels watery

Higher PVA content produces stronger glue.

2. Water

Water helps:

- * reduce thickness
- * spread the glue properly
- * reduce production cost

Too much water:

- ✗ weakens glue
- ✗ increases drying time
- ✗ reduces customer satisfaction

Always balance properly.

3. CMC

CMC is the thickener.

Its purpose:

- * thickens the glue
- * improves texture
- * prevents watery appearance
- * improves stability

Important:

CMC must be added slowly into water.

If added too quickly:

- ✗ lumps form
- ✗ glue becomes rough
- ✗ uneven texture develops

4. Preservative

Preservative prevents:

- * mould growth

- * bad smell
- * spoilage
- * bacterial growth

Without preservative:
Glue can spoil within days or weeks.

SAFETY PRECAUTIONS

Always:

Wear gloves
Work in a ventilated area
Avoid inhaling powders
Keep chemicals away from children
Wash hands after production

Do not eat or drink during production.

STEP-BY-STEP MANUFACTURING PROCESS

STEP 1 — PREPARE THE WATER

Pour approximately:

- * 5 litres of clean water

into a clean bucket.

Use room temperature water.

Dirty water can:

- * reduce quality
- * introduce bacteria
- * shorten shelf life

STEP 2 — ADD THE CMC

Slowly sprinkle:

- * 150 g CMC

while stirring continuously.

IMPORTANT:

Do NOT pour the whole CMC at once.

If you dump it:

- ✗ lumps form immediately
- ✗ mixture becomes rough
- ✗ difficult to correct later

STEP 3 — HYDRATION PROCESS

After mixing:

leave the CMC solution for:

* 2 to 4 hours

This process is called:

HYDRATION

During hydration:

- * CMC absorbs water
- * swelling occurs
- * thickness develops

After hydration:

the solution becomes:

smooth

thicker

gel-like

This is normal.

STEP 4 — ADD PVA BINDER

Add:

* 8 litres PVA binder

slowly into the hydrated CMC solution.

Stir continuously.

At this stage:

the actual glue structure begins forming.

Mix until:
smooth
even
no separation visible

STEP 5 — ADD REMAINING WATER

Add the remaining:

* 5 litres water

slowly while stirring.

This helps achieve the correct consistency.

The glue should not become:

✗ too watery

✗ too thick

Target:
smooth flowing glue.

STEP 6 — ADD PRESERVATIVE

Add:

* 30–50 ml preservative

Mix thoroughly.

This extends shelf life.

STEP 7 — OPTIONAL ADDITIVES

White Pigment

Improves:

* whiteness

* premium appearance

Fragrance

Improves:

- * customer experience
- * smell

Defoamer

Removes:

- * foam
- * excessive bubbles

STEP 8 — FINAL MIXING

Mix for:

- * 10–15 minutes

until fully uniform.

Final product should be:

- ✓ smooth
- ✓ creamy
- ✓ white
- ✓ medium thickness
- ✓ lump free

QUALITY TESTING

Before packaging:

Apply glue on:

- * paper
- * cardboard
- * wood

Allow drying.

Check:

- ✓ bonding strength
- ✓ drying speed
- ✓ smoothness
- ✓ spreading ability

COMMON PROBLEMS AND SOLUTIONS

Problem	Cause	Solution
Glue too watery	Too much water	Reduce water
Glue too thick	Too much CMC	Add little water
Lumps forming	Poor CMC mixing	Sprinkle slowly
Bad smell	No preservative	Add preservative
Weak bonding	Low PVA	Increase PVA
Too many bubbles	Overmixing	Add defoamer

PACKAGING

Recommended packaging sizes:

Size	Target Market
100 ml	Students
250 ml	Schools
500 ml	Offices
1 litre	Businesses

LABELING IDEAS

Your label should include:

- ✓ Product name
- ✓ Directions for use
- ✓ Manufacturer name
- ✓ Contact details
- ✓ Batch number
- ✓ Safety warning

Example:

> “Keep away from children.”

STORAGE

Store:

- ✓ in cool place
- ✓ away from sunlight
- ✓ tightly closed

Do not expose to:

- ✗ excessive heat
- ✗ dirty environments

Properly stored glue can last several months.

BUSINESS IDEAS

You can supply:

- * schools
- * stationery shops
- * offices
- * furniture makers
- * craft businesses
- * packaging companies

This can become:

home industry
school supply business
wholesale business

COST REDUCTION STRATEGY

To reduce production cost:

- * slightly reduce PVA
- * slightly increase water
- * balance thickness using CMC

WARNING:

Too much dilution causes:

- ✗ weak glue
- ✗ customer complaints
- ✗ poor reputation

Always maintain acceptable quality.

PREMIUM STRONG GLUE VERSION

For stronger glue:

Ingredient	Quantity	
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PVA Binder	10–12 litres
Water	7–8 litres
CMC	150–180 g

This version provides:

- ✓ stronger bonding
- ✓ better wood adhesion
- ✓ premium quality

FINAL ADVICE FROM ARACHIS ONLINE TRAINING

Success in manufacturing comes from:

consistency
cleanliness
proper measurements
patience during mixing
good packaging
customer trust

Do not rush production.

Good quality products build repeat customers and long-term profit.

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